

JIERUI LI

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Education

University of Washington

Expected Jun 2029

Ph.D. in Mechanical Engineering | GPA: 4.00/4.00

Seattle, WA

- **Research Focus:** AI for materials discovery, molecular generative modeling, molecular dynamics
- **Fellowships:** College of Engineering Dean's Fellowship; Mechanical Engineering Endowed Students First Fellowship

Carnegie Mellon University

May 2025

M.S. in Mechanical Engineering | GPA: 3.96/4.00

Pittsburgh, PA

University of Pittsburgh

May 2023

B.S. in Mechanical Engineering | GPA: 3.90/4.00

Pittsburgh, PA

Sichuan University

May 2023

B.E. in Mechanical Engineering | GPA: 3.51/4.00

Chengdu, China

- Joint 2+2 dual-degree program with the University of Pittsburgh

Projects

MacroFlow: Agentic Vitrimer Discovery

Sep 2025 – Present

Graduate Research Assistant | Vashisth Research Lab

University of Washington

- Developed a grammar-constrained Graph Transformer generative flow network for target- T_g -conditioned vitrimer design in a low-data setting, deriving the molecular grammar from 100 labeled acid-epoxide pairs and training the generation policy with trajectory balance across free, fixed-acid, and fixed-epoxide design modes.
- Fine-tuned the generator using Direct Preference Optimization and integrated RDKit validation with an XGBoost T_g surrogate; achieved 84.8–98.0% validity and 98.5–100% novelty across multiple targets from 300 to 450 K, producing top-20 shortlists with surrogate target MAEs of 3.5–9.4 K.
- Built FastAPI and Model Context Protocol services and prototyped a scientific-agent workflow that integrates molecular generation with literature-grounded candidate screening and experimental prioritization, designed for integration with Microsoft Research's CLIO framework.

EUV Photoresist Molecular Layer Deposition

Sep 2025 – Present

Graduate Research Assistant | Vashisth Research Lab

University of Washington

- Performed accelerated ReaxFF molecular dynamics simulations of molecular layer deposition reactions between trimethylaluminum and four diol precursors on the HPC cluster to compare precursor-dependent deposition behavior.
- Developed AMS trajectory-analysis workflows to quantify reaction conversion, surface-bond orientation, radial distribution functions, and deposited-layer structure across the four precursor systems.

MultiCAD: Multimodal Generative AI for CAD Automation 🔄

Mar 2025 – May 2025

Course Project | 10-623: Generative AI

Carnegie Mellon University

- Constructed 160,000+ aligned sketch-text-CAD samples by generating sketch-style images from parametric CAD models and pairing them with natural-language descriptions and construction sequences.
- Fine-tuned CLIP-ViT-B/16 with LoRA on 148,000 sketch-text pairs, improving matching accuracy from 5.2% to 21.6% on a 17,000-sample validation set.
- Developed a cross-attention Transformer decoder for autoregressive CAD sequence generation; evaluated against DeepCAD and prompted Qwen2.5-VL and GPT-4o baselines, achieving 40% lower mean Chamfer distance than DeepCAD.

Reinforcement Learning for Grade-Aware Robotic Limestone Mining

Sep 2023 – May 2025

Graduate Research Assistant | Computational Engineering & Robotics Lab

Carnegie Mellon University

- Developed a PPO-based actor-critic planner with action masking and reward shaping to jointly maximize high-grade limestone recovery and minimize autonomous wheel-loader travel distance.
- Evaluated flattened, MLP, CNN, and Transformer pile-surface representations across nine configurations (30–140 blocks) using 100,000 randomized environments per configuration.
- Reduced average planning cost by 7.74% and 4.26% versus greedy and best-first search baselines, respectively; generated full action sequences in at most 0.64 seconds, while best-first search required up to 1,980 seconds and failed on large configurations.

Experience

Carnegie Mellon University

Sep 2024 – May 2025

Teaching Assistant | AI/ML for Engineers; Deep Learning for Engineers

Pittsburgh, PA

- Led weekly office hours for 100+ graduate students in Deep Learning for Engineers, providing guidance on PyTorch implementations of CNNs, VAEs, GANs, diffusion models, and Transformer-based language models.
- Evaluated assignments and exams across AI/ML for Engineers and Deep Learning for Engineers, providing feedback on model implementation, debugging, and performance analysis.

ENN Energy Holdings Limited

Jul 2023 – Aug 2023

Test Engineer Intern – Mechanical

Langfang, China

- Conducted rotor-balancing tests on micro gas turbine assemblies and calculated permissible residual unbalance and required correction masses across multiple balance-quality grades.
- Performed vibration spectrum analysis on micro gas turbine engines to identify dominant frequency components and assess vibration characteristics.

Publications

Yuhei Sugano, **Jierui Li**, and Kenji Shimada. “Material Grade-Based Task Planning for Robotic Limestone Mining.” *Advanced Robotics*, vol. 40, no. 2, pp. 63–82, 2026. [DOI](#)

Technical Skills

Programming: Python, C++, Java, MATLAB

Machine Learning: PyTorch, scikit-learn, Hugging Face Transformers, XGBoost, NumPy, SciPy, OpenCV

Simulation & Modeling: RDKit, LAMMPS, HOOMD-blue, AMS/PLAMS, SOLIDWORKS, CATIA, Ansys

Software & HPC: FastAPI, Model Context Protocol (MCP), Git, Linux, Slurm